POLITECNICO DI MILANO

Scuola di Ingegneria Industriale e dell'Informazione

Master Degree in Mechanical Engineering
Course of Advanced Dynamics of mechanical systems

REPORT ASSIGNMENT 1A

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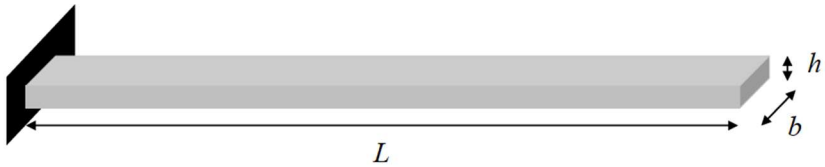
Introduction and overview of the problem

This assignment has the goal to analyze the mechanical behavior of a system deepening especially its vibrational response: this report shows how to compute its natural frequencies and mode shapes, how to study its forced response by computing the Frequency Response Functions for assigned input and output positions, and lastly how to make them representative of experimental data.

In particular, the system taken into consideration is a cantilever beam made of aluminum and having a rectangular cross-section. It has been studied by making reference to the standing wave solution of a slender beam in bending vibration.



Parameter	symbol	unit	value
Lenght	L	mm	1200
Thickness	h	mm	8
Width	b	mm	40
Density	ρ	kg/m^3	2700
Young's Modulus	E	GPa	68

Fig. 1 System and its properties

Natural frequencies and mode shapes

By considering the form of the standing wave solution:

$$w(x, t) = [A*\cos(\gamma x) + B*\sin(\gamma x) + C*\mathrm{Cosh}(\gamma x) + D*\mathrm{Sinh}(\gamma x)]*\cos(\omega t + \varphi)$$

and imposing the boundary conditions:

$$Left\ end: \begin{cases} w(0,t) = 0 \\ \left.\frac{\partial w}{\partial x}\right|_{x=0} = 0 \end{cases} \qquad Right\ end: \begin{cases} \left.\frac{\partial^2 w}{\partial^2 x}\right|_{x=L} = 0 \\ \left.\frac{\partial^3 w}{\partial^3 x}\right|_{x=L} = 0 \end{cases}$$

it is possible to obtain the system:

$$\begin{cases} A + C = 0 \\ B + D = 0 \\ A * \sin(\gamma L) - B * \cos(\gamma L) + C * \sinh(\gamma L) + D * \cosh(\gamma L) = 0 \\ -A * \cos(\gamma L) - B * \sin(\gamma L) + C * \cosh(\gamma L) + D * \sinh(\gamma L) = 0 \end{cases}$$

equivalent, in the matrix form, to the system:

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \sin(\gamma L) & -\cos(\gamma L) & \sinh(\gamma L) & \cosh(\gamma L) \\ -\cos(\gamma L) & -\sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \end{bmatrix} * \begin{bmatrix} A \\ B \\ C \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Since the first matrix can be identified as [H(γ)] and $\underline{z}$ is the vector of unknowns, it is possible to obtain the formulation:

$$[\mathrm{H}(\gamma)] * \underline{z} = \underline{0}$$

At this point it is necessary to solve the characteristic equation det[H(ω)]=0 - that is non linear - to find the natural frequencies ω_i and, subsequently, the modeshapes $\Phi_i(x)$. It is relevant to notice that the analysis is being conducted in the frequency range of [0-200]Hz.

The natural frequencies can be derived by solving iteratively the equation for gamma $[\mathrm{H}(\gamma)] = 0$ using the *matlabFunction* and the function *fsolve* in an interval going from 1 - since γ=0 equals a motionless system - to the maximum value corresponding to the maximum frequency. After finding the values of gamma that make the H matrix singular, it is possible to obtain the corresponding values of the natural frequencies ($\gamma^4 = \frac{m}{E*J} * \omega^2$) . The results are [4.50; 28.22; 79.03; 154.87] and are showed in the figure below:

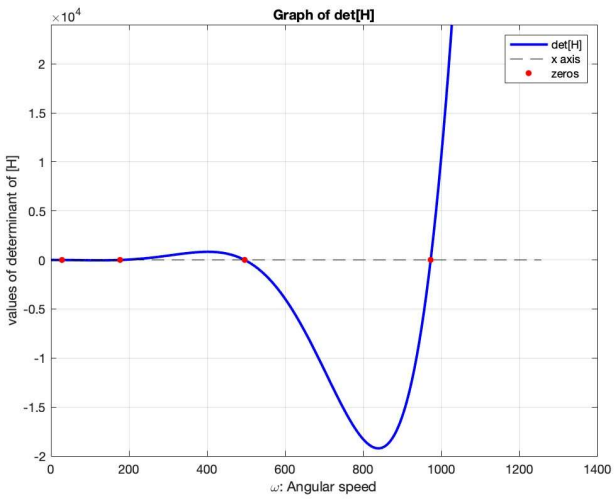


Fig. 2 Natural frequencies w_i s.t. det[H(w)]=0

Formulation of the mode-shapes

To compute and plot the mode shapes it is necessary to evaluate the unknown coefficients A, B, C and D. In order to do this, it is possible to impose arbitrarily A=1 rearranging the vector of unknowns $\underline{z}$ in the form:

$$\underline{z} = \begin{bmatrix} 1 & \underline{\hat{z}} \end{bmatrix} \text{ where } \underline{\hat{z}} = \begin{bmatrix} B & C & D \end{bmatrix}$$

In such way, the first columns of the matrix $[\mathrm{H}(\gamma)]$ - which is multiplied by the coefficient A and can be identified as $\underline{\mathrm{N}}$ - is known and the system is reduced to:

$$\begin{bmatrix} \underline{\mathrm{N}} & \left[\widehat{H}(\omega_i)\right] \end{bmatrix} * \begin{bmatrix} 1 \\ \underline{\hat{z}} \end{bmatrix} = \underline{0}, where\ \underline{\mathrm{N}}\ is\ a\ (4,1)\ column\ vector\ and\ \left[\widehat{H}(\omega_i)\right]\ is\ a\ (4,3)\ matrix$$

The system can be rewritten in order to get the unknowns and calculate the mode shapes:

$$\underline{\mathrm{N}} + \left[\widehat{H}(\omega_i)\right] * \underline{\hat{z}} = \underline{0} \rightarrow \underline{\hat{z}} = -\left[\widehat{H}(\omega_i)\right]^{-1} * \underline{\mathrm{N}}$$

$$\phi_i(x) = A * \cos(\gamma_i x) + B * \sin(\gamma_i x) + C * \cosh(\gamma_i x) + D * \sinh(\gamma_i x)$$

This procedure is calculated with the first four natural frequencies in order to obtain the first four mode shapes; the results are shown in the figure below:

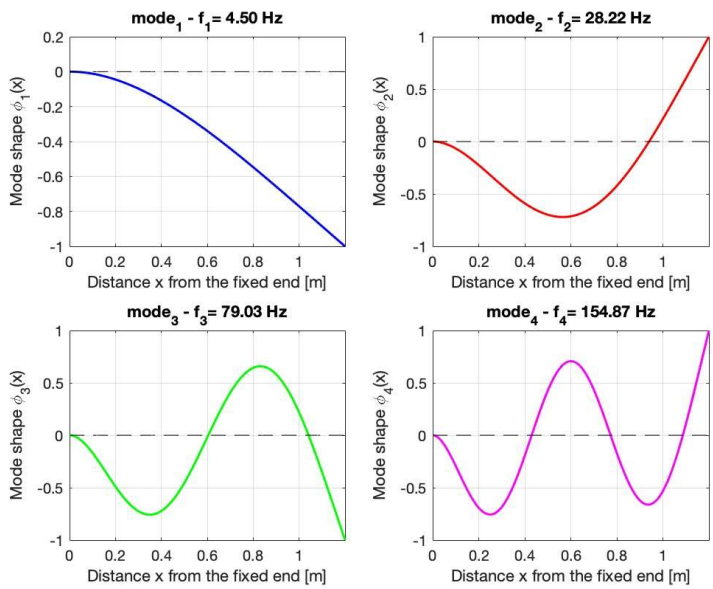


Fig. 3 First four mode shapes

For each mode it is also possible to obtain the positions of the nodes (whose number increases with higher mode shapes), which are:

- 1st mode: no nodes;
- 2nd mode: x=0.9401
- 3rd mode: x=[0.6043 1.0412]
- 4th mode: x=[0.4300 0.7729 1.0867]

Experimental FRFs

The Frequency Response Functions are expressed in the formulation:

$$G_{jk}^{EXP}(\Omega) = \sum_{i=1}^{n} \frac{\frac{\phi_i(x_j) * \phi_i(x_k)}{m_i}}{-\Omega^2 + 2j\xi_i\omega_i\Omega + \omega_i^2}$$

where x_k and x_j represent the input and output positions respectively (where the force and the sensor are located) and ω_i represent the natural frequencies. In this case, the chosen ones are:

$$\underline{x_k} = [1.2 \quad 1.0 \quad 0.8 \quad 0.3]\ in\ [m]; \quad \underline{x_j} = [0.2 \quad 0.6 \quad 0.75 \quad 0.8]\ in\ [m]$$

Moreover, a minus sign has been included in order to consider a more realistic case in which the force is going downwards, whereas the sensor measures in the upward direction.

Going on, it is necessary to define the damping values ξ_i and the modal mass m_i, which is defined as

$$m_i = \int_0^L m * \phi_i^2(x)dx$$

and has been evaluated through the *trapz* function.

At this point it is possible to plot the FRFs, as shown below.

In all these cases:

- There are four peaks representing the first four mode shapes in correspondence of the frequencies [4.50; 28.22; 79.03; 154.87];
- The peaks in the amplitude diagram are limited and do not tend to infinite, as expected for a damped system;
- The phase remains in the interval $[+\pi;\ -\pi]$;
- In the quasi-static region the phase is equal to $+\pi$ - and not zero, as it would be expected - because of the minus considered at the numerator of the $G_{jk}^{EXP}(\Omega)$ due to the unconventional versus assigned to the force (which is downwards) which lead to $\phi_i(x_j) * \phi_i(x_k) < 0$.

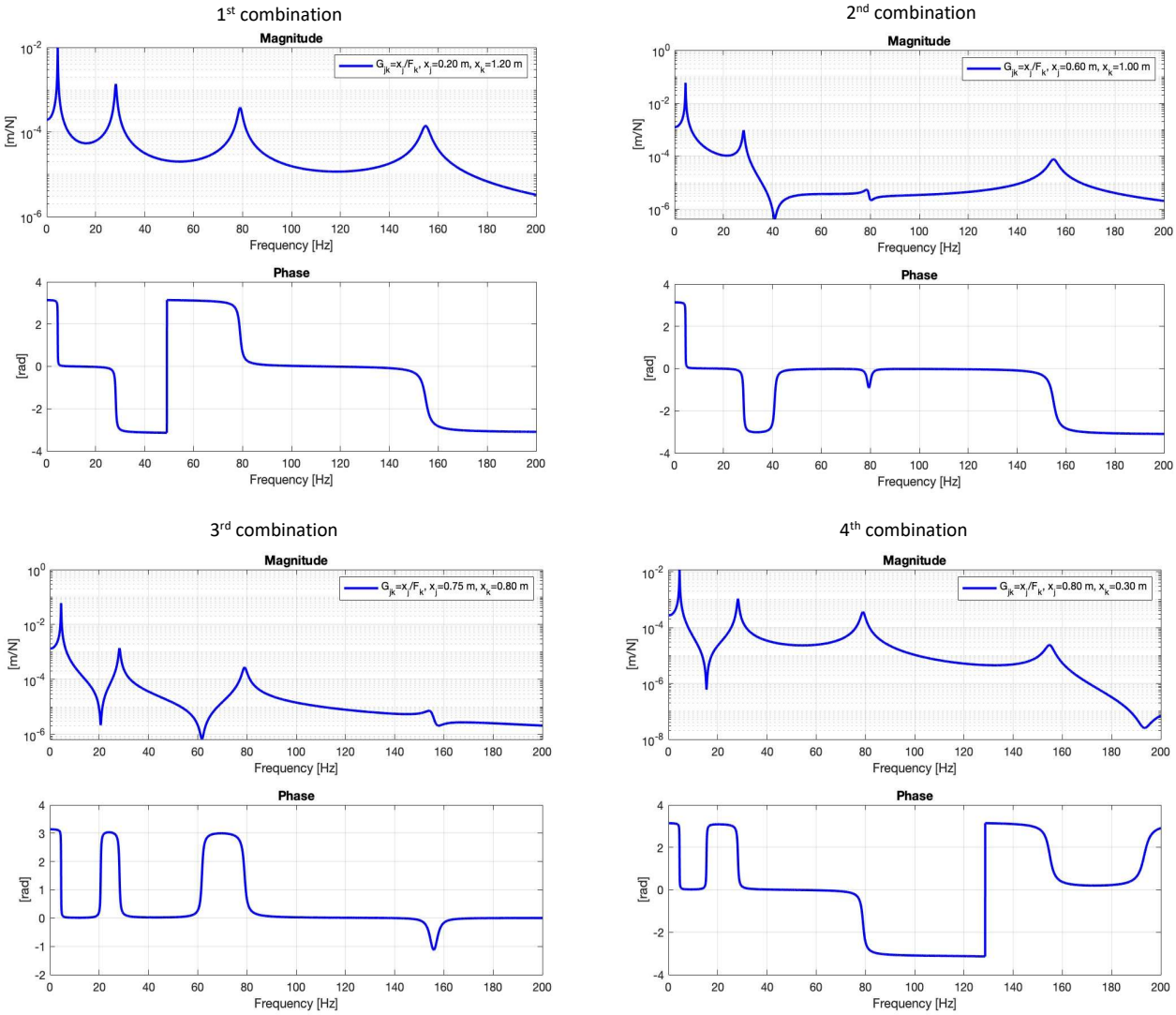


Fig. 4 Plot of the FRFs

It is also possible to plot the FRFs considering one or more nodes an inputs and/or outputs. For example, firstly, it may be considered the case of having the node of the second mode $\widehat{x_k} = 0.9401$ as input and $x_j = 0.70$ as output; then, the reciprocal case ($x_k = 0.7$; $\widehat{x_j} = 0.9401$). The resulting FRFs are shown below:

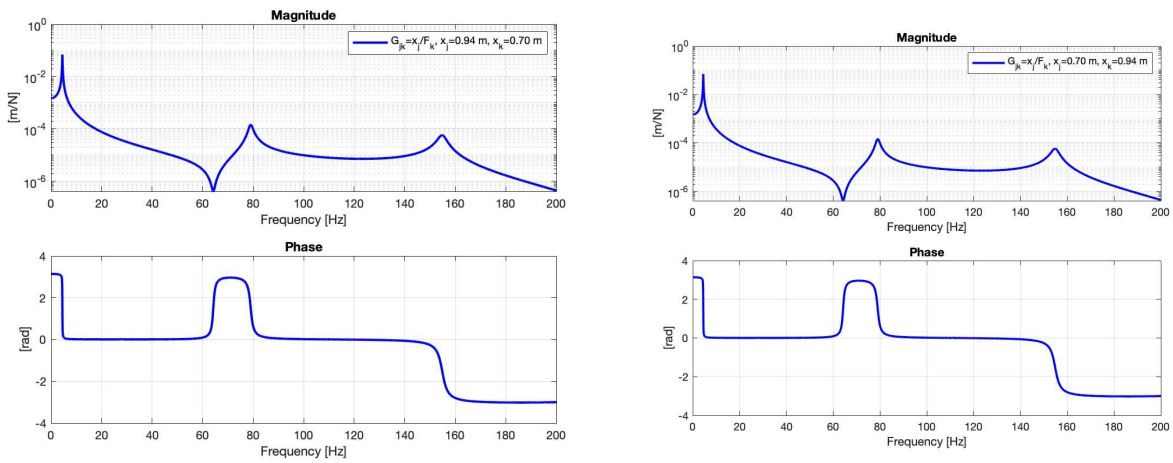


Fig. 5 FRFs with a node

It can be observed that in both the cases the peak of the second mode is absent and this mode shape gives no contribution, since its node has been considered as the input/output position. In particular, in the first case the system is forced near the node and the mode is not controllable; whereas, in the second case the response is measured near the same node and the mode is not observable. This allow also to state that inverting the positions of input and output locations - giving an example of reciprocity - the FRF remains the same.

Numerical FRFs

Model parameters identification method

If the FRFs presents above can be considered representative of an experimental test, it is now possible to compute the numerical form of the FRFs, which are represented by the function:

$$G_{jk}^{NUM}(\Omega) = \sum_{i=1}^{n} \frac{A_{jk}^{(i)}}{-\Omega^2+2j\xi_i\omega_i\Omega+\omega_i^2} + \frac{R_{jk}^L}{\Omega^2} + R_{jk}^H$$

In the function above:

- The sum represents the contribution of all the (i) modes in the frequency range of interest $[\Omega_{min};\ \Omega_{max}]$ which are resonating;
- The residuals are introduced to consider also the influence of the modes not included in the selected frequency range, in particular R_{jk}^L stands for low frequency modes (that are expected to be in the seismic region in the considered range) whereas R_{jk}^H stands for high frequency modes (that are expected to be in the quasi-static region in the considered range);
- The unknowns are: ξ_i, ω_i, $A_{jk}^{(i)}$, R_{jk}^L and R_{jk}^H, where the first three are real, whereas the last two are complex (having a real and an imaginary part).

In the case taken into consideration, it has been considered n equal to 1 meaning that in the frequency range of interest only one mode shape is studied (because there is only one resonance peak) obtaining a realistic and acceptable response - whereas theoretically each mode contributes minimally to the resonance peak.

<u>For each vibration mode</u> (corresponding to a single resonance peak), considering as many FRFs as the number of j-k pairs, the damping ratio ξ_i and the natural frequency ω_i do not change since they do not depend on the specific combination of input and output locations, whereas the other terms vary with each FRF. For this reason, if N FRFs are taken into consideration, the number of unknowns for a single mode is equal to (2+5*N) and they can be collected in a vector of parameters.

At this point it is necessary to estimate the modal parameters that best fit the experimental data relying on a least square minimization procedure, using the *lsqnonlin* function in Matlab. Considering:

- M as the the length of the frequency vector, with Ω_s varying in the range $[\Omega_{min};\ \Omega_{max}]$;
- N as the number of the j-k pairs, so to the number of FRFs;
- The expression of a single FRF:

$$G_r^{NUM}(\Omega_s) = \frac{A_r^{(i)}}{-\Omega_s^2 + 2j\xi_i\omega_i\Omega_s + \omega_i^2} + \frac{R_r^L}{\Omega_s^2} + R_r^H$$

the error function to be minimized corresponds to the squared difference between two complex numbers evaluated for all the Ω_s and all the j-k pairs:

$$\epsilon = \sum_{r=1}^{N}\sum_{s=1}^{M} Re\left(G_r^{EXP}(\Omega_s) - G_r^{NUM}(\Omega_s)\right)^2 + Im\left(G_r^{EXP}(\Omega_s) - G_r^{NUM}(\Omega_s)\right)^2.$$

It is possible to proceed by considering one mode per time and defining the initial guesses for the unknowns required by the *lsqnonlin* function:

- ω_0 has been guessed identifying graphically the resonance peak in the frequency range of interest for the considered vibration mode;
- ξ_0 has been predicted through the slope of the phase diagram and has also been checked with the half power points method:

$$\xi_0 = -\frac{1}{\omega_i * \left.\frac{\partial \phi}{\partial \Omega}\right|_{\Omega=\omega_i}};$$

- Each A_{jk0} has been guessed evaluating the imaginary part of the FRFs in correspondence of the resonance peak ($G_{jk}^{EXP}(\omega_0)$) and multiplying it for the factor:

$$A_i = -Im\left(FRF_i^{EX}(\omega_0)\right) * 2 * \omega_0^2 * \xi_0$$

- The residuals R_0^L and R_0^H have been imposed null because ideally they do not have any influence.

The vector $\underline{x_0}$ collecting the unknowns is then:

$$\underline{x_0} = [\omega_0, \xi_0, A_{jk}(1), A_{jk}(2), A_{jk}(3), A_{jk}(4), 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,]$$

and this allows to solve the minimization problem obtaining the vector of parameters requested to compute the G_r^{NUM}.

Numerical FRFs and comparison of the results

Once all the parameters and the corresponding numerical FRFs have been found, it is possible to plot the lasts and compare them with the ones previously found (the experimental FRFs) in correspondence of the resonance peaks for each j-k combination: this allows to evaluate the quality of the identification process. In the pictures below are shown the results for the four different vibration modes.

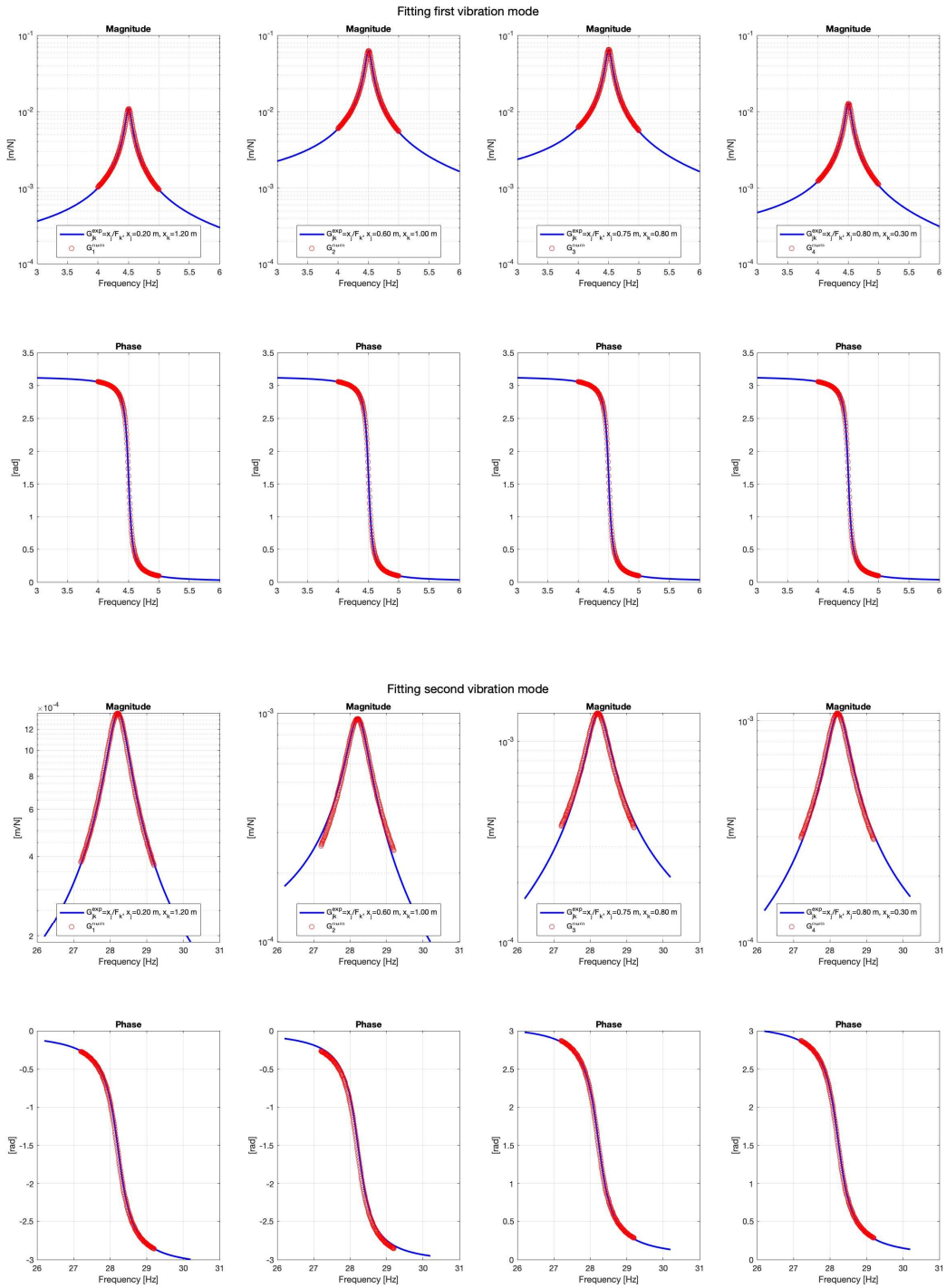


Fig.6 Numerical FRFs: 1st and 2nd modes

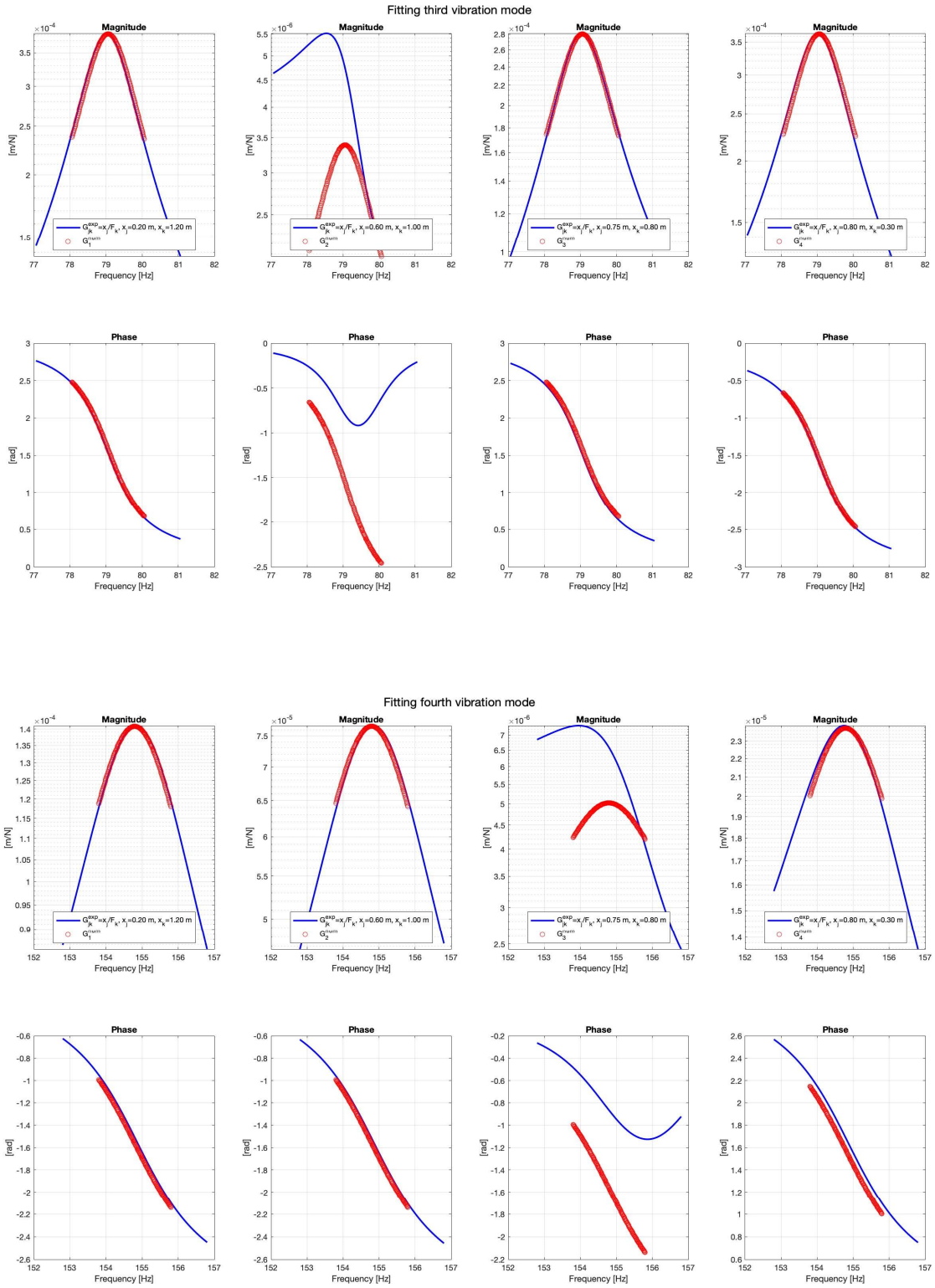


Fig. 7 Numerical FRFs: 3rd and 4th modes

It is worth noticing that in the last two modes some FRFs do not fit adequately: this can be explained observing that for the fourth and third FRFs there isn't any resonance peak in the frequency range of interest due to the proximity of a sensor to a node of the mode.

Comparison of the parameters

To conclude, it is also possible to compare the parameters of the model and the identified ones focusing, in particular, on the similarity between the simulated and identified mode shapes.

In the following figures the continuous line represents the mode shapes of the model defined above ($\phi_i^{EXP}(x) = A * \cos(\gamma_i x) + B * \sin(\gamma_i x) + C * \cosh(\gamma_i x) + D * \sinh(\gamma_i x)$), whereas the dots are the identified ones. In particular, the lasts have been derived from the definition of the parameter $A_{jk}^{(i)}$:

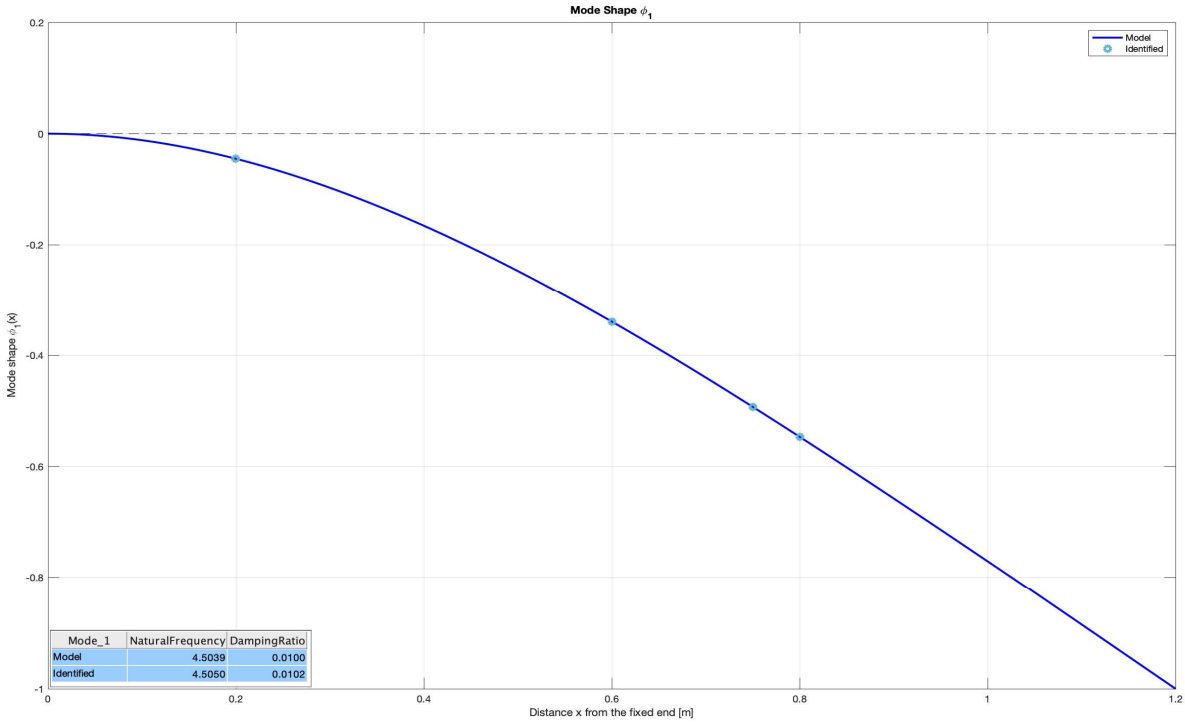
$$A_{jk} = \frac{x_j * x_k}{m_i} \rightarrow x_j = \frac{A_{jk} * m_i}{x_k}$$

and has been multiplied for a common scaling factor equal to the ratio between the experimental value of the first mode shape in a specific output location $\phi^{EXP}(x_j)$ and the numerical value evaluated in the same point $\phi_i^{NUM}(x_j)$ - obtained with the least square minimization process. Then, the results have been normalized with respect to the maximum value of the experimental mode shapes ϕ_i^{EXP}.

Observing the plots, it can be stated that there is a good agreement between the data, that indeed respect two requirements:

1. Being the result of an accurate identification process (they have the same shape);
2. Having been multiplied by a common scaling factor (the results are overlapped).

The results are shown below:



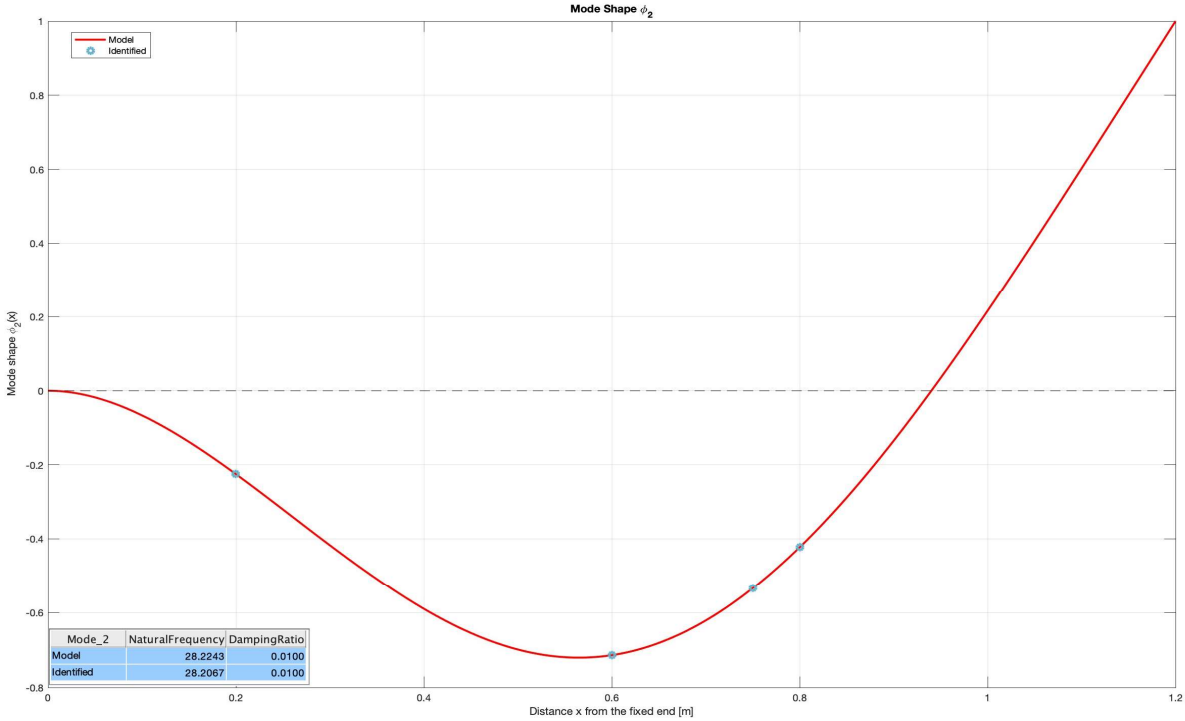
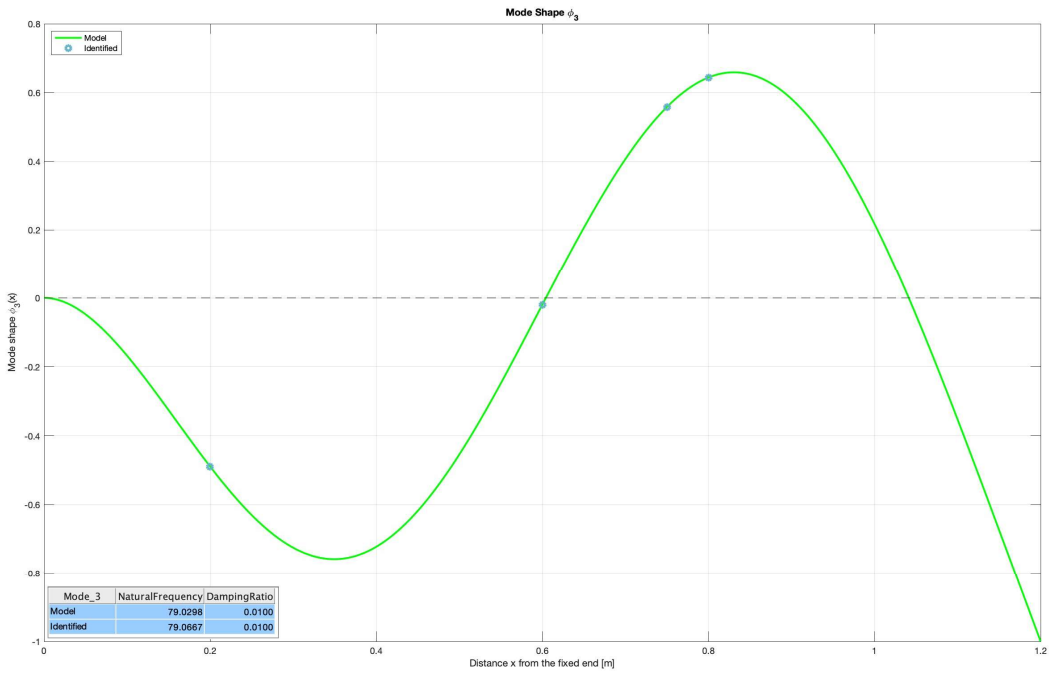


Fig. 8 Mode shapes comparison: 1st and 2nd modes

Mode_1	NaturalFrequency	DampingRatio
Model	4.5039	0.0100
Identified	4.5050	0.0102

Mode_2	NaturalFrequency	DampingRatio
Model	28.2243	0.0100
Identified	28.2067	0.0100



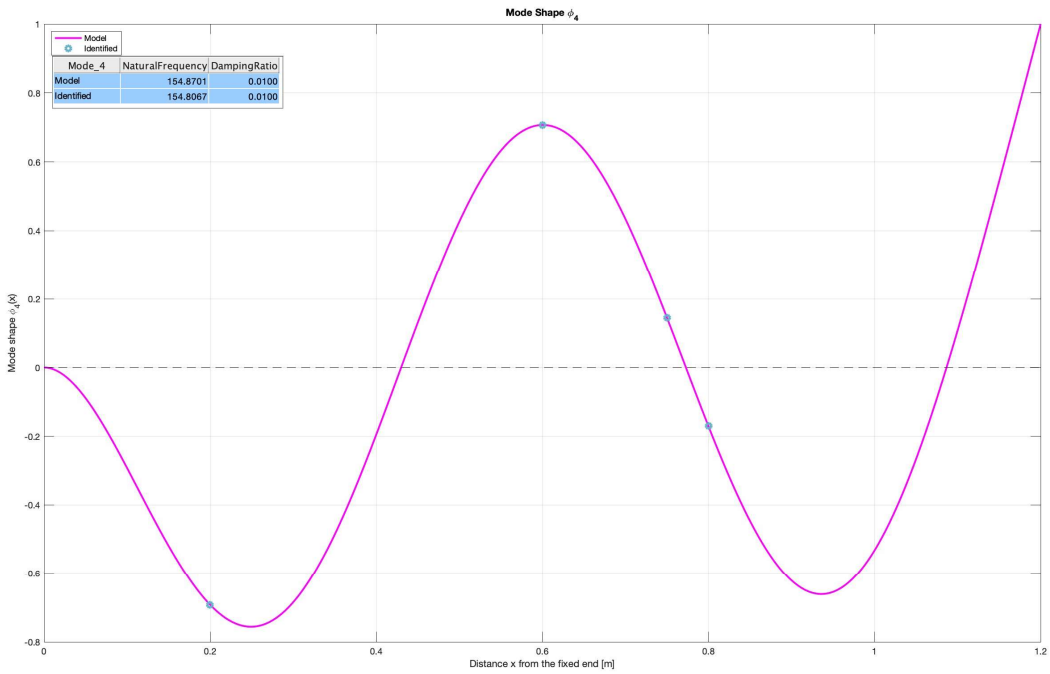


Fig. 9 Mode shapes comparison: 3rd and 4th modes

Mode_3	NaturalFrequency	DampingRatio
Model	79.0298	0.0100
Identified	79.0667	0.0100

Mode_4	NaturalFrequency	DampingRatio
Model	154.8701	0.0100
Identified	154.8067	0.0100